MIL-A-6589A(USAF)
AMENDMENT 2
26 MAY 86
SUPERSEDING
AMENDMENT 1
27 July 1978

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL, SINGLE
CHANNEL, TYPE A-2

This amendment forms a part of MIL-A-6589A(USAF), dated 13 December 1968, and is approved for use within the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

PAGE 1

* 1.1 Add title "Scope".

* Delete "2.1" and add: "2.1 Government documents".

* Add:

"2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATION, Federal delete "QQ-P-416" and its title, and add: "PPP-B-601 Boxes, Wood, Cleated Plywood".

* Under SPECIFICATION, Military delete "MIL-D-1000", "MIL-E-5272" and "MIL-C-7769". Add: "DOD-D-1000 Drawing, Engineering and Associated List" and "MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited".

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

2.1 Under STANDARDS, Military, delete: "MS33586" and substitute: "MIL-STD-889 Dissimilar Metals".

PAGE 3

3.3.3.1 Line 3, delete "MS33586" and substitute "MIL-STD-889".

AMSC N/A FSC 6605

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* Add:

"3.3.2.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

* Delete "3.7.2" and add:

"3.7.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

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* Delete "4.6.7" and add:

"4.6.7 Low temperature. The amplifier shall be subjected to low temperature tests. The equipment shall be placed within the test chamber and cooled to and maintained at a temperature of -65°C (-85°F) for a period of 48 hours. At the end of the 48 hour period, and while still at this low temperature, the amplifier shall be operated throughout its range without exhibiting erratic characteristics. The temperature shall then be changed to -55°C (-65°F) and maintained for an additional 24-hour period or until stabilization is reached, whichever is longer. At the end of the 24-hour period and while still at the -55°C temperature, the amplifier shall be subjected to the individual tests stipulated for low temperature operation and the same tolerances shall apply. There shall be no damage which would adversely affect subsequent operation of the amplifer."

* Delete "4.6.8" and add:

"4.6.8 Altitude. The amplifier shall be subjected to the following altitude tests. The equipment shall be placed within the test chamber and the absolute internal pressure reduced to 3.44 inches of mercury (corresponding to an altitude of 50,000 feet above sea level), at a temperature of -54 ± 2°C for a continuous period of 4 hours. While at this temperature and pressure, the amplifier shall be subjected to the individual tests stipulated for low temperature operation and the same tolerances shall apply. There shall be no damage which would adversely affect subsequent operation of the amplifier."

* Delete "4.6.9" and add:

"4.6.9 High temperature. The amplifier shall be subjected to the following high temperature testing: The equipment shall be placed within the chamber and the internal temperature of chamber raised to 71°C (160°F) with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat shall not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The item of equipment shall be maintained at 71°C (160°F) for a period of 50 hours, and while still at this temperature, the equipment shall be examined,

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tested, and operated. The amplifier shall operate without indication of erratic action or damage. After the chamber has been returned to room temperature, for a minimum period of 4 hours, the amplifier shall meet the individual test requirements specified herein and there shall be no damage which would adversely affect subsequent operation of the amplifier."

PAGE 14

* Delete "4.6.11" and add:

"4.6.11 Vibration. The amplifier shall be subjected to the following vibration testing:"

* Add:

"4.6.11.1 Test configuration. This testing shall be conducted while the amplifier is being operated on normal power input and a signal level of 0.4V. The amplifier shall be mounted on apparatus dynamically similar to reciprocating engine powered aircraft and to gas turbine powered aircraft, in turn, and shall be functioning in accordance with the provisions of this specification during the entire test period, when practical. At the end of the test period, the test specimen shall be inspected thoroughly for damage or defects resulting from the vibration tests. The amplitude or acceleration for the frequency cycling test shall be within ± 10 percent of the specified values. Vibration test shall be conducted under both resonant and cycling conditions according to the Vibration Test Schedule (Table I)."

TABLE I

VIBRATION TEST SCHEDULE (Times shown refer to one axis of vibration)			
Type	Vibration at Room Temp.	Vibration at 160°F(71°)	Vibration at -65°F(55°C)
Resonance	60 minutes	15 minutes	15 minutes
Cycling	60 minutes	15 minutes	15 minutes

Add:

"4.6.11.2 Resonance. Resonant frequencies of the test specimen shall be determined by varying the frequency of applied vibration slowly through the specified range of frequencies at vibratory accelerations not exceeding those shown in Figure 3. This procedure shall be followed successively for vibration applied along three mutually perpendicular axes of the test specimen. Whenever practicable, functioning of the test specimen shall be checked concurrently with the operation of scanning the frequency range for resonant frequencies. The test specimen shall be vibrated at the indicated resonant conditions for the periods shown in the Vibration Test Schedule and with the applied double amplitude or vibratory acceleration specifed in Figure 3. These periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration

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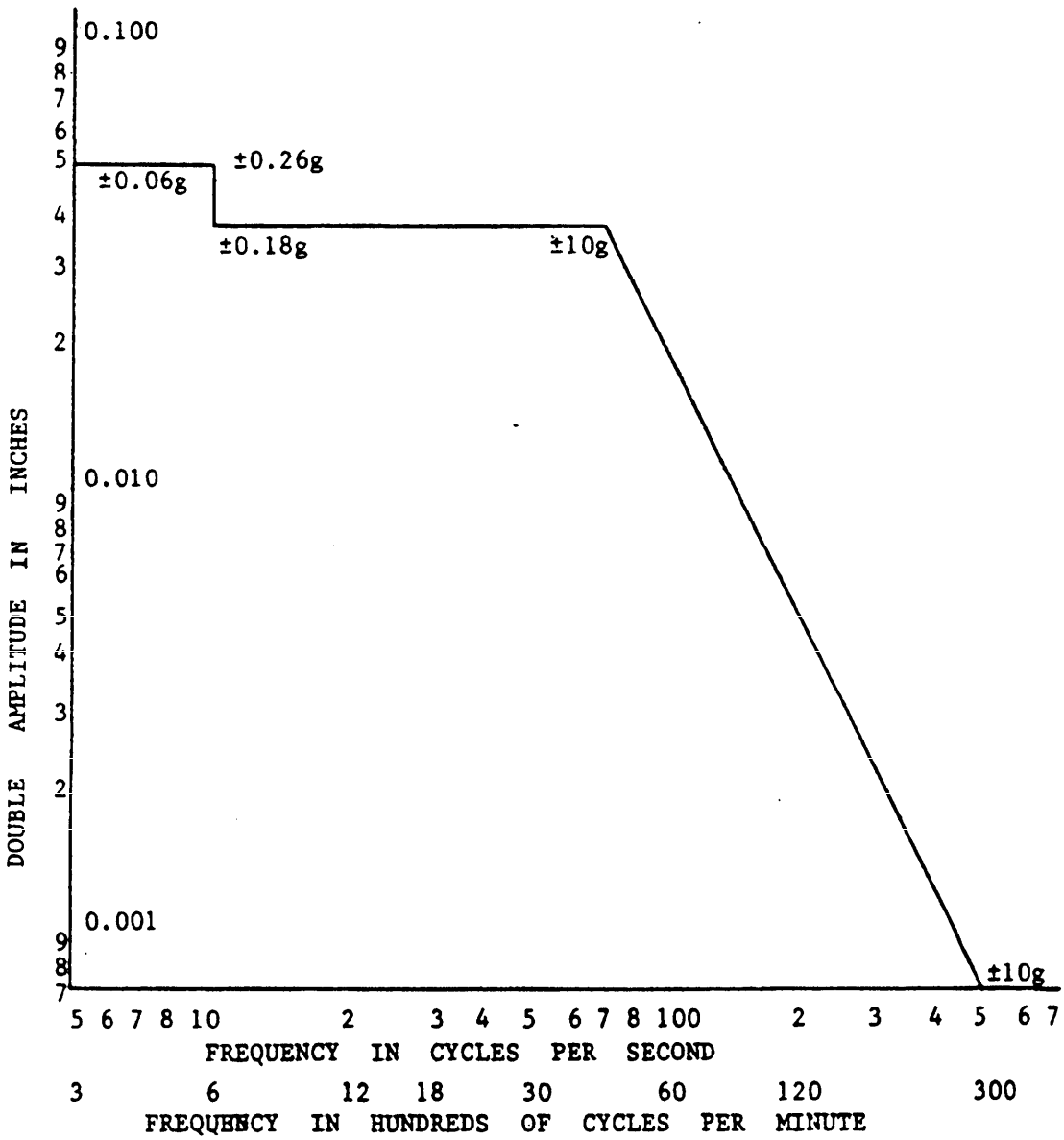


FIGURE 3. Range curve for vibration tests.

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applied along any one axis, the test period may be accomplished at the most severe resonance or the period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. When resonant frequencies are not apparent within the specified frequency range, the specimen shall be vibrated for periods twice as long as those shown for resonance in the Vibration Test Schedule at a frequency of 55 cps and an applied double amplitude of 0.060 inch."

* Add:

"4.6.11.3 Cycling. A vibration test shall be conducted with a constant applied vibratory accleration of ± 10 g with the frequency cycling between 10 and 500 cps in 15-minute cycles at an applied double amplitude of 0.036 inch or an applied acceleration of ± 10 g, whichever is the limiting values. Vibration shall be applied along each of three mutually perpendicular axes according to the Vibration Test Schedule (Table I)."

* Add:

"4.6.11.4 Alternate tests. In those instances where facilities for conducting the above tests are not available, and with the approval of the contracting agency, the test shall be conducted at room temperature. That portion of the test applying to resonant conditions shall be the same except that the time of vibration shall be three hours in each direction along three mutally perpendicular axis of the test specimen. The cycling portion shall be accomplished between 10 and 55 cps only. Time of vibration shall be one hour in each direction along three mutually perpendicular axes."

* Add:

"4.6.11.5 Post vibration. After the vibration period, the amplifier shall meet the requirements stipulated under individual tests. In addition, no looseness in the mechanism nor damage to any part of the amplifier shall result from this test."

Delete "4.6.12" and add:

"4.6.12 Humidity. The amplifier shall be subjected to the following humidity tests: The equipment shall be placed in a test chamber which is capable of being sealed, and the temperature and relative humidity raised to 71°C (160°F) and 95 percent, respectively, during a 2-hour period. The source of heat for the chamber shall be so arranged that radiant heat shall not fall upon the specimen. The temperature of 71°C (160°F) and relative humidity of 95 percent shall be maintained for a period of six hours. At the conclusion of the 6-hour period the heat shall be shut off. During the following 16-hour period the temperature must drop at a uniform rate with condensation to 38°C (100°F) or less. The cycle shall be repeated a sufficient number of times to the extend the total of the test to 360 hours (15 cycles). At the conclusion of the 360 hour period the equipment shall be examined, tested, and operated. The amplifier shall meet the requirements specified for individual tests. There shall be no deterioration as a result of this test which could in any manner prevent the amplifier from meeting operational requirements during its normal service life. Distilled or demineralized water having a ph value of between 6.8 and 7.2 at 25°C (77°F) shall be used to obtain the desired humidity."

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* Delete "4.6.13: and add:

"4.6.13 Fungus. The amplifier shall be subjected to the fungus test described below. The unit selected shall not have previously been subjected to the Salt Spray Test."

Add:

"4.6.13.1 Tests. At least five fungi shall be used in each test. Five groups of fungi are listed below and one type of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a ph value between 6.8 and 7.2 at 25°C (77°F)) shall be prepared by sterilization in convenient containers introduced directly into a stock culture of one fungus and shaken vigorously so that a well sporulated suspension will result without disturbing the agar. This process shall be repeated for each type fungus. The separate spore suspensions from the 5 types of fungi shall then be mixed together in an atomizer to provide a composite suspension. The equipment, including applicable external connections shall be placed in a mold chamber maintaining an internal temperature of 30°C (86°F) and a relative humidity of 95 percent, and sprayed with the suspension of mixed spores similar to those encountered in tropical climates. The test period shall be 28 days. At the conclusion of the test, the equipment shall be examined, tested, and operated. The amplifier shall then meet the requirements specified for individual tests. There shall be no deterioration as a result of this specified for individual test which could in any manner prevent the amplifier from meeting operational requirements during its normal service life."

Add:

"4.6.13.2 Organisms.

Group I	Chastomium globosum USDA 1042.2 or Myrothecium verrucaria USDA 1334.2
Group II	Rhisopus nigricans S.N. 32 or Apergillus niger USDA TC215-4247
Group III	Aspergillus flavus AMC No. 26 or Aspergillus terreus PQMD 82J
Group IV	Penicillium luteum USDA 1336.1, Penicillium sp. USDA 1336.2 or Pencillium citrinum ATCC 9848
Group V	Memnoniella echinata AMC No. 37 or Fusarium moniliforme USDA 1004.1"

Add:

"4.6.13.3 Stock Culture Designation and Source.

Symbols	Source
USDA	U.S Department of Agriculture, Beltsville, Maryland
ATCC	American Type Culture Collection, Georgetown University, Washington, D.C.
PQMD	Philadelphia Q.M. Depot, 2800 South 20th Street, Philadelphia, Pennsylvania

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AFLC Air Force Logistics Command, Materials Laboratory, Wright-Patterson Air Force Base, Dayton, Ohio

S.N. Dr. W.B. Weston, Biological Laboratories, Harvard University, Cambridge, Massachusetts"

* Delete "4.6.14" and add:

"4.6.14 Salt spray. The amplifier shall be subjected to the salt spray tests specified herein."

Add:

"4.6.14.1 Apparatus. Apparatus used in the salt spray test shall include the following:

Exposure chamber with racks for supporting specimens.
Salt solution reservior.
Means for atomizing salt solution, including suitable nozzles and compressed air supply.
Chamber heating means and control.
Means for humidifying the air at a temperature above the chamber temperature."

* Add:

"4.6.14.1.1 Chamber. The chamber and all accessories shall be made of material which will not affect the corrosiveness of the fog, such as glass, hard rubber, plastic, or wood other than plywood. In addition, all parts which come in contact with test items shall be of materials that will not cause electrolytic corrosion. The chamber and accessories shall be so constructed and arranged that there is no direct impinging of the spray or dripping of the condensate on the test items, so that no liquid which has come in contact with the specimens returns to the salt solution reservoir. The chamber shall be properly vented."

* Add:

"4.6.14.1.2 Atomizers. The atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog."

* Add:

"4.6.14.1.3 Air supply. The compressed air entering the atomizers shall be free from all impurities such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely divided dens fog with the atomizer or atomizers used. To insure against clogging the atomizers by salt deposition the air should have a relative humidity of at least 85 percent at the point of release from the nozzle. A satisfactory method is to pass the air in very fine bubbles through a tower containing heated water. The temperature of the water should be 35°C (95°F) and often higher. The permissible temperature increases with volume of air and with decreasing heat insulation of the chamber and temperature of its surroundings. It should not exceed a value above which an

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excess of moisture is introduced into the chamber (for example, 43°C (109°F), at an air pressure of 12 lbs. per square inch), or value which makes it impossible to meet the requirements for operating temperuature."

* Add:

"4.6.14.1.5 Salt solution. The salt used shall be sodium chloride containing on the dry basis not more than 0.1 percent of sodium iodide and not more than 0.2 percent of total impurities. The solution shall be prepared by dissolving 20 plus or minus 2 parts by weight of salt in 80 parts by weight of distilled or other water containing not more than 200 parts per million of total solids. The solution shall be kept free from solids by filtration or decantation. The solution shall be adjusted to and maintained at a specific gravity of from 1.126 to 1.157 and at a ph of between 6.5 to 7.2 when measured at a temperature between 33° and 36°C (92° and 97°F). Only C. P. hydrochloric acid or C. P. Sodium hydroxide shall be used to adjust the ph. The ph measurement shall be made electrometrically using a glass electrode with a saturated potassium chloride bridge or by a colorimetric method such as bromothymol blue provided the results are equivalent to those obtained with the electrometric method."

* Add:

"4.6.14.2 Operating conditions.

4.6.14.2.1 Temperature. The test shall be conducted with a temperature in the exposure zone maintained at 35°C. Satisfactory methods for controlling the temperature accurately are by housing the apparatus in a properly controlled constant temperature room, by throughly insulating the apparatus and preheating the air to the proper temperature prior to atomization and by jacketing the apparatus and controlling the temperature of the water or of the air used. The use of immersion heaters for the purpose of maintaining the temperature within the chamber is prohibited."

* Add:

"4.6.14.2.2 Atomization. The conditions maintained in all parts of the exposure zone shall be such that a suitable receptacle placed at any point in the exposure zone will collect form 0.5 to 3 ml. of solution per hour for each 80 sq. cm. of horizontal collecting area (10 cm. diameter) based on an average of at least 16 hours. The solution collected shall have a sodium chloride content of from 18 to 22 percent (sp. gr. of from 1.126 to 1.157 when measured at a temperature between 33° and 36°C). At least two clean fog collecting receptacles shall be used, one placed nearest to any nozzle and one farthest from all nozzles. Receptacles shall be fastened so that no drops of solution from specimens or other sources will be collected. When using nozzles made of material nonreactive to the salt solution, suitable atomization has been obtained in boxes having a volume of less than 12 cubic feet with the following conditions:

a. Nozzle pressure of from 12 to 18 pounds per square inch.
b. Orifices of from 0.02 to 0.03 inch in diameter.
c. Atomization of approximately 3 quarts of salt solution per 10 cubic feet of box volume per 24 hours.

When using large-size boxes having a volume considerably in excess of 12 cubic feet the above conditions may have to be modified in order to meet the requirements for operating conditions."

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* Add;

"4.6.14.2.3 Preparation of equipment. The equipment shall be mounted in the chamber and electrical and/or mechanical connections completed where operation is required during the test period. Access covers and inspection plates shall be in place except when such covers would normally be removed for service use. When operation is not required external connections may be plugged to simulate actual service conditions."

* Add:

"4.6.14.3 Length of test. The length of the salt spray test shall be not less than 50 hours. At the end of the test period, the amplifier shall be subjected to the individual test specified herein and the same tolerances shall apply. However, the amplifier may be washed in clear running water prior to conducting operational tests. The amplifier shall also be subjected to visual examination for corrosion. There shall be no corrosion or deterioration as a result of this test, which could in any manner prevent the amplifier from meeting operational requirements during its normal service life."

Delete paragraph 5 through 5.2 in their entirety and substitute the following:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Amplifier shall be cleaned in accordance with C-1 of MIL-P-116.

5.1.1.2 Drying. Amplifier shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each amplifier shall be preserved and packaged in quantity unit packs of one each in accordance with Method IIB of MIL-P-116. Each unit shall be wrapped with polyfilm, L-P-378 or equal, and placed within a form fitting primary container. After primary container has been bagged to meet the requirements of Method IIB, apply sufficient cushioning material between primary and unit container, of a type, density, and thickness to insure shock transmission does not exceed peak values in Gs established for the item, when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Amplifiers shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Style A or B, class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with protection required.

5.2.2 Level B. Amplifiers packaged as specified in 5.1.1 shall be packaged in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level."

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6.2b Delete and substitute:

"b. Level of packaging and packing required (see 5.)."

* Delete "6.3".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to last previous amendment.

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